***4.** Suppose that m is square-free, and that k and $\bar{k}$ are positive integers such that $k\bar{k} \equiv 1 \pmod{\phi(m)}$. Show that $a^{k\bar{k}} \equiv a \pmod{m}$ for all integers a. (H)

- $k\bar{k} = \phi(m)\,l + 1 \qquad (l \in \mathbb{N})$

- $m = p_1 \cdots p_i\, q_1 \cdots q_j$

  s.t ① p's and q's are prime (square-free)

  ② $d = p_1 \cdots p_i = (a, m)$

- $(a^{\phi(m)})^l \cdot \cancel{a} \equiv \cancel{a} \pmod{\cancel{m}}$

Note that $(a/d,\ q_1 \cdots q_j) = 1$

iff $(a^{\phi(m)})^l\, (a/d) \equiv (a/d) \pmod{q_1 \cdots q_j}$

iff $(a^{\phi(m)})^l \equiv 1 \pmod{q_1 \cdots q_j}$

Note that $(a/d,\ q_1 \cdots q_j) = 1$

---

since $(a, q_1 \cdots, q_j) = 1$

we have $a^{\phi(m)} \equiv \left(a^{\phi(q_1 \cdots q_j)}\right)^{\phi(p_1 \cdots p_i)}$

$\equiv 1^{\phi(p_1 \cdots p_j)}$

$\equiv 1 \pmod{q_1 \cdots, q_j}$ by Euler.

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***5.** Suppose that m is a positive integer that is not square-free. Show that there exist integers a_1 and a_2 such that $a_1 \not\equiv a_2 \pmod{m}$, but $a_1^k \equiv a_2^k \pmod{m}$ for all integers $k > 1$.

- let $m = p^2 q$ (p is prime)
- let $a_1 = pq$, $a_2 = p^2 q$
- then $\underset{pq}{\underset{|||}{a_1}} \not\equiv \underset{0}{\underset{|||}{a_2}} \pmod{m}$ $\because$ $p^2 q \nmid pq$

  $p^2 q \mid p^2 q$
- if $k \geq 2$ then $p^2 q \mid (pq)^k = p^2 q^2 (pq)^{k-2}$ therefore $(p^2 q)^k \equiv (pq)^k \equiv 0 \pmod{m}$

  $p^2 q \mid (p^2 q)^k$

2. Solve $x^5 + x^4 + 1 \equiv 0 \pmod{3^4}$. $\Rightarrow$ No solution

- $x^5 + x^4 + 1 \equiv 0 \pmod 3$ iff $x \equiv 1 \pmod 3$

- $x^5 + x^4 + 1 \equiv 0 \pmod 9$

if $x \equiv 1 \pmod 9$ then $x^5 + x^4 + 1 \equiv 3 \pmod 9$

if $x \equiv 4 \pmod 9$ then
$$\begin{aligned} x^5 + x^4 + 1 &\equiv x^4(x+1) + 1 \\ &\equiv 5^4 \cdot (5) + 1 \\ &\equiv 5^5 + 1 \\ &\equiv 5^6 \cdot (5-1) + 1 \quad \text{(Euler)} \\ &\equiv 2 + 1 \equiv 3 \end{aligned}$$

if $x \equiv 7 \pmod 9$ then
$$\begin{aligned} x^5 + x^4 + 1 &\equiv (-2)^5 + (-2)^4 + 1 \\ &\equiv -32 + 16 + 1 \\ &\equiv 3 \pmod 9 \end{aligned}$$

Note. by Hensel

$$\begin{aligned} f(x) &\equiv f(1) + f'(1)(x-1) \quad \text{(Taylor)} \\ &\quad + f''(1)/2!\,(x-1)^2 \\ &\quad \cdots \\ &\equiv f(1) + f'(1)(x-1) \\ &\equiv 3 + 9(x-1) \\ &\equiv 3 \pmod 9 \end{aligned}$$

$\because$ if $x \equiv 1 \pmod 3$ then $9 \mid (x-1)^2, (x-1)^3 \ldots$

7. Solve $x^3 + x^2 - 5 \equiv 0 \pmod{7^3}$.

- $x^3 + x^2 - 5 \equiv 0 \pmod 7$ iff $x \equiv 2 \pmod 7$

- $x^2(x+1)$

$$\equiv (y+2)^2 (y+3) \qquad \begin{pmatrix} x = 7k+2 \\ y = 7k \end{pmatrix}$$

$$\equiv (y^2+4y+4)(y+3)$$

$$\equiv \cancel{y^3} + \cancel{7y^2} + 16y + 12$$

$$\equiv \quad 112k + 12 \equiv 5 \pmod{7^3}$$

$$112k \equiv -7 \pmod{7^3}$$

$$16k \equiv -1 \pmod{7^2}$$

$$\cancel{16}k \equiv \cancel{48}\ 3 \pmod{7^2}$$

then $x \equiv 7k+2 \pmod{7^3}$

$\equiv 23 \pmod{7^3}$

Note. by Hensel $f(x) = x^3 + x^2 - 5$

$f(2) = 8 + 4 - 5 = 7$

$f'(2) = 12 + 4 = 16$

$f''(2) = 14$

if $x \equiv 2 \pmod 7$

$$f(x) \equiv f(2) + f'(2)(x-2) + f''(2)/2!\ (x-2)^2$$

$$\equiv 7 + 16(x-2) + 7(x-2)^2$$

$$\equiv 16x - 25 \pmod{7^3}$$

$$x = \frac{343+25}{16} = 23$$

9. Suppose that $f(a) \equiv 0 \pmod{p^j}$ and that $f'(a) \not\equiv 0 \pmod{p}$. Let $\overline{f'(a)}$ be an integer chosen so that $f'(a)\overline{f'(a)} \equiv 1 \pmod{p^{2j}}$, and put $b = a - f(a)\overline{f'(a)}$. Show that $f(b) \equiv 0 \pmod{p^{2j}}$.

- $f(a) + f'(a)(b-a)$

$$\equiv f(a) + f'(a)\left(-f(a)\overline{f'(a)}\right)$$

$$\equiv f(a) - f(a) \equiv 0 \pmod{p^{2j}}$$

- if $k \geq 2$ $\quad f(a)^2 \mid (b-a)^k = \left(-f(a)\overline{f'(a)}\right)^k$

with $p^j \mid f(a)$ we get $(b-a)^k \equiv 0 \pmod{p^{2j}}$

- Taylor $\quad f(b) \equiv f(a) + f'(a)(b-a)$

$$+ f''(a)/2!\,(b-a)^2$$

$$+ f'''(a)/3!\,(b-a)^3 \quad \cdots$$

$$\equiv 0 + 0 + 0 + \cdots$$

Note. $f^{(n)}(a)/n!$ is integer

- $f(x)$를 왼쪽으로 $a \in \mathbb{Z}$ 만큼 평행이동 $\Rightarrow g(x)$

- then $\quad g(x) = f(a) + f'(a)x$

$$+ f''(a)/2!\,x^2$$

$$+ f'''(a)/3!\,x^3 + \cdots$$

- 왼쪽으로 정수만큼 평행이동하면 계수는 당연히 정수다.

Example 13 Discuss the solutions of $x^2 + x + 223 \equiv 0 \pmod{3^j}$.

$4(x^2+x+223) \equiv 0 \pmod{3^j}$

$(2x+1)^2 \equiv -891 \pmod{3^j}$

let $y = 2x+1$ then $y^2 \equiv -11 \cdot 9^2 \pmod{3^j}$

$j=1$: $y^2 \equiv 0 \pmod 3$. $y \equiv 0 \pmod 3$

$j=2$: $y^2 \equiv 0 \pmod 9$ $y \equiv 0, 3, 6 \pmod 9$

$j=3$: $y^2 \equiv 0 \pmod{27}$ $y \equiv 0, 9, 18 \pmod{27}$

$j=4$: $y^2 \equiv 0 \pmod{81}$ $y \equiv 0, 9, 18, 27, 36, 45, 54, 63, 72 \pmod{81}$

$j \geq 5$ $y = 9z$ then $81z^2 \equiv -11 \cdot 81 \pmod{3^j}$

$z^2 \equiv -11 \pmod{3^{j-4}}$

$j=5$ then $z^2 \equiv 1 \pmod 3$ let $f(z) = z^2+11$

$z \equiv 1, 2 \pmod 3$ $f'(z) = 2z$

then
$$\begin{cases} f(1) \equiv 0 \pmod 3 \\ f(2) \equiv 0 \pmod 3 \end{cases}$$

$$\begin{cases} f'(1) \equiv 2 \not\equiv 0 \pmod 3 \\ f'(2) \equiv 1 \not\equiv 0 \pmod 3 \end{cases}$$

by Hensel $z^2 \equiv -11 \pmod{3^{j-4}}$ 의 해는 2개로 유일하다.

Hensel's Lemma.

$$\begin{cases} f(r_k) \equiv 0 \pmod{p^k} \\ f'(r_k) \not\equiv 0 \pmod{p} \end{cases}$$ then unique r_{k+1} in modulo p^{k+1}

s.t $$\begin{cases} f(r_{k+1}) \equiv 0 \pmod{p^{k+1}} \\ r_{k+1} \equiv r_k \pmod{p^k} \end{cases}$$

$\because$ let $r_{k+1} = r_k + t \cdot 3^k$

$$f(r_{k+1}) \equiv \cancel{f(r_k)}^{\,0} + f'(r_k)(x - r_k) + f''(r_k)/2!\,\cancel{(x-r_k)^2}^{\,0} + \cdots$$

$$\equiv f'(r_k)(t \cdot 3^k) \equiv 0 \pmod{3^{k+1}}$$

$$\Longleftrightarrow \quad f'(r_k) \cdot \underbrace{t}_{\text{unique solution exists.}} \equiv 0 \pmod{3}$$

t is unique $\pmod 3$

$r_{k+1} = r_k + t \cdot 3^k$ is unique $\pmod{3^{k+1}}$

---

z가 $\pmod{3^{j-4}}$에서 2개의 해를 가지므로.

$y = 9z$는 $\pmod{3^{j-2}}$에서 2개의 해를 가지고.

$\pmod{3^j}$에서 18개의 해를 가진다.

$x \equiv (y-1) \cdot 2^{-1} \pmod{3^j}$는 18개의 해를 가진다.

2. Prove that $2x^3 + 5x^2 + 6x + 1 \equiv 0 \pmod 7$ has three solutions by use of Theorem 2.29.

iff

Theorem 2.29 *The congruence $f(x) \equiv 0 \pmod p$ of degree n, with leading coefficient $a_n = 1$, has n solutions if and only if $f(x)$ is a factor of $x^p - x$ modulo p, that is, if and only if $x^p - x = f(x)q(x) + ps(x)$, where $q(x)$ and $s(x)$ have integral coefficients, $q(x)$ has degree $p - n$ and leading coefficient 1, and where either $s(x)$ is a polynomial of degree less than n or $s(x)$ is zero.*

$$4(2x^3 + 5x^2 + 6x + 1)$$

$$\equiv 8x^3 + 20x^2 + 24x + 4$$

$$\equiv \underbrace{x^3 - x^2 + 3x + 4 \equiv 0}_{} \pmod 7$$

since $(x^3 - x^2 + 3x + 4)(x^4 + x^3 - 2x^2 - 2x) \equiv x^7 - x \pmod 7$

this equation has 3 solutions

---

4. Prove that if $f(x) \equiv 0 \pmod p$ has j solutions $x \equiv a_1, x \equiv a_2, \cdots, x \equiv a_j \pmod p$, there is a polynomial $q(x)$ such that $f(x) \equiv (x - a_1)(x - a_2) \cdots (x - a_j)q(x) \pmod p$. (H)

induction

if $f(a) \equiv 0$ then $f(x) \equiv \cancel{f(a)}^{\,0} + f'(a)(x-a)$ (Taylor)

$$+ f''(a)/2!\,(x-a)^2$$

$$\cdots$$

$$\equiv (x-a)\left\{ f'(a) + \frac{f''(a)}{2!}(x-a) + \cdots \right\} \pmod p$$

---

suppose $f(a_1) \equiv \cdots \equiv f(a_j) \equiv 0$

with $f(x) \equiv (x-a_1)\cdots(x-a_j)g(x) \pmod p$

we know $f(a_{j+1}) \equiv \underbrace{(a_{j+1} - a_1)}_{\text{reduced residue}} \cdots \underbrace{(a_{j+1} - a_j)}_{\text{reduced residue.}} g(a_{j+1}) \equiv 0 \pmod p$

we know

$(a_{j+1} - a_1, p)$

$= \cdots = (a_{j+1} - a_j, p) = 1$

therefore $g(a_{j+1}) \equiv 0$

$$\underbrace{g(x) = (x - a_{j+1})\,r(x)}_{\text{귀납적 가정에 의하여 성립}} \pmod p$$

6. Let m be composite. Prove that Theorem 2.26 is false if "mod p" is replaced by "mod m."

Theorem 2.26 *The congruence $f(x) \equiv 0 \pmod{p}$ of degree n has at most n solutions.*

$2x \equiv 0 \pmod{4}$

has solution $x \equiv 0 \pmod{2}$

i.e $x \equiv 0, 2 \pmod{4}$

---

7. Show that if the prime number p in Theorem 2.28 is replaced by a composite number m then the statement becomes false.

Theorem 2.28 *If $F(x)$ is a function that maps residue classes (mod p) to residue classes (mod p), then there is a polynomial $f(x)$ with integral coefficients and degree at most $p-1$ such that $F(x) \equiv f(x) \pmod{p}$ for all residue classes x (mod p).*

$\{0, 1, 2, 3\} \pmod{4}$

let $f(x) = Ax^3 + Bx^2 + Cx + D$

then $f(0) \equiv D \pmod{4}$

$f(2) \equiv 8A + 4B + 2C + D$

$\equiv 2C + D \pmod{4}$

$f(0)$과 $f(2)$는 중복되므로 다르게 만들 수 없다.

10. Write $1/1 + 1/2 + \cdots + 1/(p-1) = a/b$ with $(a, b) = 1$. Show that $p^2 | a$ if $p \geqslant 5$.

---

since $p \nmid (p-1)!$

we can check if $p^2 \mid \sum_{k=1}^{p-1} \frac{(p-1)!}{k}$ $(p \geq 5)$

$$\sum_{k=1}^{p-1} (p-1)! \left(\frac{1}{k}\right) = \sum_{k=1}^{(p-1)/2} (p-1)! \left(\frac{1}{k} + \frac{1}{p-k}\right)$$

$$= \sum_{k=1}^{(p-1)/2} (p-1)! \frac{p}{k(p-k)}$$

---

① $$\sum_{k=1}^{(p-1)/2} \frac{1}{k(p-k)} \equiv -\sum_{k=1}^{(p-1)/2} \frac{1}{k^2} \pmod{p}$$

② $$\underbrace{\sum_{k=1}^{p-1}}_{\text{reduced residue}} \frac{1}{k^2} \equiv \underbrace{\sum_{k=1}^{p-1}}_{\text{reduced residue}} k^2 \equiv \frac{(p-1)\,p\,(2p-1)}{6} \equiv 0 \pmod{p}$$

③ $$2 \sum_{k=1}^{(p-1)/2} \frac{1}{k^2} \equiv \sum_{k=1}^{p-1} \frac{1}{k^2} \equiv 0 \pmod{p}$$

therefore $$\sum_{k=1}^{(p-1)/2} \frac{1}{k(p-k)} \equiv 0 \pmod{p}$$